Practice assignment 9 - Not Graded

This assignment will not be graded and is only for practice.
Multiple Select Question (MSQ)
1 point
Consider a function $f: R^2 \to R$ f : R2 → R defined as:

$$f(x,y) = \begin{cases} \frac{2xy}{\sqrt{2(x^2+y^2)}} & \text{if } (x,y) \neq (0,0) \\ 1 & \text{if } (x,y) = (0,0). \end{cases}$$

f(x, y) = { 2(x2+y2) √ 2xy1if (x, y) = (0, 0)if (x, y) = (0, 0).

Choose the set of correct options.

- [ ] $|x| \leq \sqrt{x^2+y^2}$ |x| ≤ x2 + y2 √ and $|y| \leq \sqrt{x^2+y^2}$ |y| ≤ x2 + y2 √ .
- [ ] If $(x,y) \to (0,0)$ (x, y) → (0, 0), then $f(x,y) \to 0$ f(x, y) → 0.
- [ ] $\lim_{(x,y)\to(0,0)} f(x,y)$ (x,y)→(0,0)lim f(x, y) exists.
- [ ] $\lim_{(x,y)\to(0,0)} f(x,y) = 0$ (x,y)→(0,0)lim f(x, y) = 0.
- [ ] $f(x,y)$ f(x, y) is continuous at the origin (0,0).

No, the answer is incorrect.
Score: 0
Accepted Answers:
$|x| \leq \sqrt{x^2+y^2}$ |x| ≤ x2 + y2 √ and $|y| \leq \sqrt{x^2+y^2}$ |y| ≤ x2 + y2 √ .
If $(x,y) \to (0,0)$ (x, y) → (0, 0), then $f(x,y) \to 0$ f(x, y) → 0.
$\lim_{(x,y)\to(0,0)} f(x,y)$ (x,y)→(0,0)lim f(x, y) exists.
$\lim_{(x,y)\to(0,0)} f(x,y) = 0$ (x,y)→(0,0)lim f(x, y) = 0.

1 point
Consider the following functions

$$f(x,y) = (\sqrt{1-x}, \sqrt{y})$$

f(x, y) = (1 − x √

, y
√
)

$$g(x, y) = \sin((x^2 - 1)^2 + y^4)$$ g(x, y) = sin ((x2 − 1)2 + y4)

$$h(x, y) = \frac{x}{x^2 + y^2}$$ h(x, y) = x2+y2x

$$s(x, y) = x^2 e^{y^2}$$ s(x, y) = x2ey2

Which of the following is true?

- [ ] Domain of $f = \{(x, y) \mid x \leq 1, y \geq 0\}$ f = {(x, y) ∣ x ≤ 1, y ≥ 0}
- [ ] Domain of $h = R^2$ h = R2
- [ ] $g \circ f(x, y) = \sin(x^2 - 1 + y^2)$ g ∘ f(x, y) = sin (x2 − 1 + y2)
- [ ] $((g \circ f) \times h)(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}$ ((g ∘ f) × h)(x, y) = x2+y2sin (x2+y2)
- [ ] $\lim_{(x, y) \to (0, 0)} ((g \circ f) \times h + s)(x, y) = 0$ (x,y)→(0,0)lim((g ∘ f) × h + s)(x, y) = 0

No, the answer is incorrect.
Score: 0
Accepted Answers:

Domain of $f = \{(x, y) \mid x \leq 1, y \geq 0\}$ f = {(x, y) ∣ x ≤ 1, y ≥ 0}

$\lim_{(x, y) \to (0, 0)} ((g \circ f) \times h + s)(x, y) = 0$ (x,y)→(0,0)lim((g ∘ f) × h + s)(x, y) = 0

1 point

Match the functions of two variables in Column A with their graphs given in Column B.

- [ ] i → → 4, ii → → 3
- [ ] i → → 3, ii → → 4
- [ ] iii → → 1, iv → → 2
- [ ] iii → → 2, iv → → 1

No, the answer is incorrect.
Score: 0
Accepted Answers:

i → → 3, ii → → 4

iii → → 1, iv → → 2

1 point

Consider two functions $f : R^2 \to R^2$ and $g : R^2 \to R$ f : R2 → R2 and g : R2 → R defined as:

$$f(x, y) = (x^2, y^2)$$ f(x, y) = (x2, y2)

and

$$g(x,y)=\begin{cases}\frac{x^2-y^2}{x^2+y^2} & \text{if } (x,y)\neq(0,0)\\ 0 & \text{if } (x,y)=(0,0)\end{cases}$$
g(x, y) = {x2+y2x2−y20if (x, y) = (0, 0)if (x, y) = (0, 0)

Which of the following option(s) is (are) true?

☐ $g\circ f(x,y)=\begin{cases}\frac{x^2-y^2}{x^2+y^2} & \text{if } (x,y)\neq(0,0)\\ 0 & \text{if } (x,y)=(0,0)\end{cases}$ g ∘ f(x, y) = {x2+y2x2−y20if (x, y) = (0, 0)if (x, y) = (0, 0)

☐ $g\circ f(x,y)=\begin{cases}\frac{x^4-y^4}{x^4+y^4} & \text{if } (x,y)\neq(0,0)\\ 0 & \text{if } (x,y)=(0,0)\end{cases}$ g ∘ f(x, y) = {x4+y4x4−y40if (x, y) = (0, 0)if (x, y) = (0, 0)

☐ $\lim_{(x,y)\to(0,0)} g(x,y)$ (x,y)→(0,0)lim g(x, y) exists.

☐ $\lim_{(x,y)\to(0,0)} g\circ f(x,y)$ (x,y)→(0,0)lim g ∘ f(x, y) exists.

☐ $\lim_{x\to 0}\lim_{y\to 0} g\circ f(x,y)=\lim_{y\to 0}\lim_{x\to 0} g\circ f(x,y)$ x→0lim y→0lim g ∘ f(x, y) = y→0lim x→0lim g ∘ f(x, y)

☐ $\lim_{x\to 0}\lim_{y\to 0} g(x,y)=1$ x→0lim y→0lim g(x, y) = 1

☐ $\lim_{y\to 0}\lim_{x\to 0} g\circ f(x,y)=-1$ y→0lim x→0lim g ∘ f(x, y) = −1

☐ $\lim_{(x,y)\to(1,1)} g(x,y)$ (x,y)→(1,1)lim g(x, y) does not exist.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$g\circ f(x,y)=\begin{cases}\frac{x^4-y^4}{x^4+y^4} & \text{if } (x,y)\neq(0,0)\\ 0 & \text{if } (x,y)=(0,0)\end{cases}$ g ∘ f(x, y) = {x4+y4x4−y40if (x, y) = (0, 0)if (x, y) = (0, 0)

$\lim_{x\to 0}\lim_{y\to 0} g(x,y)=1$ x→0lim y→0lim g(x, y) = 1

$\lim_{y\to 0}\lim_{x\to 0} g\circ f(x,y)=-1$ y→0lim x→0lim g ∘ f(x, y) = −1

Numerical Answer Type (NAT)
Consider a function $f(x,y)=xy$ f(x, y) = xy. If the directional derivative of f f at point (a,b) (a, b), in direction of (1,-1) and (1,1) are 1 and 5 respectively. Then find the value $\frac{2a+b}{\sqrt{2}}$ 2 √ 2a+b.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 7
1 point
Consider the following statements:

Statement 1: Let $f: R^2 \to R$ be a function such that $\lim_{x \to a} \lim_{y \to b} f$ exists, where $a, b \in R$. Then $\lim_{(x,y) \to (a,b)} f$ exists.

Statement 2: Let $f: R^2 \to R$ be a function such that $\lim_{y \to b} \lim_{x \to a} f = f(a,b)$, where $a, b \in R$. Then f is continuous.

Statement 3: Let $f: R^2 \to R$ be a function such that $\lim_{x \to a} \lim_{y \to b} f = \lim_{y \to b} \lim_{x \to a} f = f(a,b)$, where $a, b \in R$. Then f is continuous.

Statement 4: Let $f: R^2 \to R$ be a function such that $f_x(a,b)$ exists. Then f is continuous.

Statement 5: Let $f: R^2 \to R$ be a function such that $f_y(a,b)$ exists. Then f is continuous.

How many statements are correct?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: String) 0
1 point

Consider the following two functions $f_1, f_2: R^2 \to R$ defined as:

$$f_1(x,y) = xy + 2x$$
$$f_2(x,y) = x^2 + 2y + 3x^2 y.$$

Define a matrix A as follows:

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x}(3,0) & \frac{\partial f_2}{\partial x}(1,1) \\ \frac{\partial f_1}{\partial y}(3,2) & \frac{\partial f_2}{\partial y}\left(\sqrt{\frac{7}{3}}, 2\right) \end{bmatrix}.$$

What will be the value of $det(A)$?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -6
1 point

Comprehension Type Question

An earthworm is crawling about on a sunny day. We draw coordinates axes on the ground where it is crawling and find that the body temperature of the earthworm at a point (x,y) is given by the function:

$$T(x,y) = 2x^2 + 3xy + y^2$$

Use this information to answer questions 8,9, and 10.

1 point

Which of the following option(s) is (are) true?

☐ $\lim\limits_{(x,y)\to(1,1)} T(x,y) = \lim\limits_{x\to 1}\lim\limits_{y\to 1} T(x,y)$

☐ $T(x,y)$ is continuous at (1,1)

☐ $T(x,y)$ is a linear function.

☐ $T(cx,cy) = c\,T(x,y)$.

☐ If the earthworm approaches the point $(1,2)$ from any direction, then the body temperature of the earthworm approaches 12 units.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\lim\limits_{(x,y)\to(1,1)} T(x,y) = \lim\limits_{x\to 1}\lim\limits_{y\to 1} T(x,y)$

$T(x,y)$ is continuous at (1,1)

If the earthworm approaches the point $(1,2)$ from any direction, then the body temperature of the earthworm approaches 12 units.

If the gradient of $T(x,y)$ is (25, 18) at a point, then find the body temperature of the earthworm at that point.

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 77

1 point

1 point

Which of the following option(s) is (are) true?

☐ If the earthworm moves parallel to the $X-$axis, then rate of change of its body temperature at the point (a,b) is given by $4a+3b$.

☐ If the earthworm moves parallel to the $Y-$axis, then rate of change of its body temperature at the point (a,b) is given by $2a+3b$.

☐ If the earthworm moves parallel to the straight line joining the origin and the point $(2,3)$, then rate of change of its body temperature at the point $(1,1)$ is given by $\frac{29}{\sqrt{13}}$.

☐ If the earthworm moves parallel to the straight line joining the origin and the point $(0,1)$, then rate of change of its body temperature at the point $(1,2)$ is given by 10.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If the earthworm moves parallel to the $X-$axis, then rate of change of its body temperature at the point (a, b) is given by $4a + 3b$.

If the earthworm moves parallel to the straight line joining the origin and the point $(2, 3)$, then rate of change of its body temperature at the point $(1, 1)$ is given by $\frac{29}{\sqrt{13}}$.

Check Answers